

MAT 320 HW08

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Due 11/1

Problem 1. Let (X, d) be a metric space and let E be a subset of X . Show that if $U \subseteq X$ is open then $U \cap E = \emptyset$ if and only if $U \cap \overline{E} = \emptyset$.

Solution. If $U \cap \overline{E} = \emptyset$ then $U \cap E = \emptyset$ is trivial because $E \subseteq \overline{E}$, as we showed in last week's homework.

Conversely, suppose for the sake of contradiction that $U \cap E = \emptyset$ but $U \cap \overline{E}$ is nonempty. Let x be in $U \cap \overline{E}$. Since x is a limit point of E , we can find a sequence (x_n) of points in E whose limit is x . In particular, for every $\varepsilon > 0$, $x_n \in B(x, \varepsilon)$ eventually. Because U is open, we can pick ε such that $B(x, \varepsilon) \subseteq U$. However, there exists an n such that $x_n \in B(x, \varepsilon)$, which means x_n is in both U and E ; this contradicts $U \cap E = \emptyset$, so $U \cap \overline{E}$ is empty. $\square$

Problem 2. Let (X, d) be a metric space. Show that (X, d) is compact if and only if for any collection of closed sets $\{F_i \subseteq X : i \in I\}$ such that $\bigcap_{i \in I} F_i = \emptyset$, there is a finite subcollection $\{F_i \subseteq X : i \in J\}$ such that $\bigcap_{i \in J} F_i = \emptyset$.

Solution. Let $\mathcal{U} = \{U_i : i \in I\}$ be any open cover of X . Letting $F_i = X \setminus U_i$, we see that

$$\bigcap_{i \in I} F_i = \bigcap_{i \in I} X \setminus U_i = X \setminus \bigcup_{i \in I} U_i = \emptyset.$$

This means there exists a finite set of indices $i_1, \dots, i_n$ such that

$$\emptyset = \bigcap_{k=1}^n F_{i_k} = \bigcap_{k=1}^n X \setminus U_{i_k} = X \setminus \bigcup_{k=1}^n U_{i_k} \iff X = \bigcup_{k=1}^n U_{i_k}.$$

That is, given any open cover $\mathcal{U}$ of X , we have extracted a finite collection of $\mathcal{U}$ (namely $\{U_{i_k} : 1 \leq k \leq n\}$) that is an open subcover, which is exactly what it means for X to be compact. $\square$

Problem 3. Prove directly from the definition of compact set that the set

$$K = \{0\} \cup \{1/n \in \mathbb{R} : n \in \mathbb{N}\}$$

is a compact subset of $\mathbb{R}$.

Solution. Let $\mathcal{U} = \{U_i : i \in I\}$ be an open cover of X . One of these U_i must contain 0; call this set U . By definition of openness, there exists an $\varepsilon > 0$ such that $B(0, \varepsilon) \subseteq U$. Let N an integer such that $\varepsilon > \frac{1}{N}$. If $n < N$, then $|\frac{1}{n} - 0| < \varepsilon$, meaning $\frac{1}{n} \in B(0, \varepsilon) \subseteq U$. If $n \geq N$, then there exist indices $i_1, \dots, i_N$ such that U_{i_k} contains $\frac{1}{k}$ because each element of K is contained in at least one open set in $\mathcal{U}$. These two cases encompass all positive integers, so we have found $N + 1$ (i.e. finitely many) sets in $\mathcal{U}$ such that

$$U \cup U_{i_1} \cup U_{i_2} \cup \dots \cup U_{i_N} \supset K.$$

That is, we have extracted a finite subcover of $\mathcal{U}$, hence K is compact. $\square$

Problem 4. Show that $(0, 1)$ is not compact by showing an example of an open cover admitting no open subcover. Do you know other ways of showing that $(0, 1)$ is not compact? Challenge: find as many different proofs as possible.

Solution 1. Consider the open cover

$$\mathcal{U} = \{U_\alpha = (0, \alpha) : \alpha \in (0, 1)\}.$$

Then, given any finite collection of indices $\alpha_1, \alpha_2, \dots, \alpha_n \in (0, 1)$ that are WLOG ordered such that $\alpha_1 \leq \dots \leq \alpha_n$, we get

$$U_{\alpha_1} \cup \dots \cup U_{\alpha_n} = (0, \alpha_n)$$

because $U_\alpha \subseteq U_\beta$ if $\alpha \leq \beta$. However, this union is clearly not equal to $(0, 1)$, so this finite subcollection of $\mathcal{U}$ is not a cover of $(0, 1)$. $\square$

Solution 2. We showed in class that compact sets are always closed. However, $(0, 1)$ is not closed, so it is not compact. $\square$

Solution 3. Alternatively, we showed in class that a subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded. Again, $(0, 1)$ is not closed, so it cannot be compact. $\square$

Solution 4. Since compactness implies sequential compactness, we can prove that $(0, 1)$ is not compact by proving that it is not sequentially compact. Indeed, the sequence $x_n = \frac{1}{n}$ for $n > 1$ converges to 0 in $\mathbb{R}$, so any subsequence also converges to 0 in $\mathbb{R}$. Therefore, no subsequence of (x_n) converges in $(0, 1)$, so $(0, 1)$ is not sequentially compact, hence not compact. $\square$

Problem 5. Show that if $C \subseteq \mathbb{R}^n$ is bounded then $K = \overline{C}$ is compact.

Solution. We note that $\overline{C}$ is bounded:

Claim 5.1 — $\text{diam } C = \text{diam } \overline{C}$.

Proof. Since $C \subseteq \overline{C}$,

$$\text{diam } C \leq \text{diam } \overline{C}.$$

To show the opposite inequality, we prove equivalently that

$$\text{diam } \overline{C} \leq \text{diam } C + \varepsilon$$

for every $\varepsilon > 0$. Indeed, let $x, y \in \overline{C}$. Then, x and y are limit points of C , so we can find $x', y' \in C$ such that $d(x, x')$ and $d(y, y')$ are both less than $\frac{\varepsilon}{2}$. The triangle inequality then says that

$$d(x, y) \leq d(x, x') + d(x', y') + d(y', y) < \frac{\varepsilon}{2} + \text{diam } C + \frac{\varepsilon}{2} = \text{diam } C + \varepsilon.$$

Thus $\text{diam } C + \varepsilon$ is an upper bound for $d(x, y)$, meaning it is at least as large as $\text{diam } C$, as desired. $\square$

Since $\text{diam } C$ is finite, so is $\text{diam } \overline{C}$, hence $\overline{C}$ is bounded. Moreover, $\overline{C}$ is closed, as shown in the previous homework. Thus, $\overline{C}$ a closed and bounded subset of $\mathbb{R}^n$, so, as we proved in class, it is compact. $\square$

Problem 6. A function $f: X \rightarrow Y$ between metric spaces is called a **Lipschitz** function if there exists a constant $L > 0$ such that

$$d_Y(f(x), f(y)) < L \cdot d_X(x, y)$$

for all $x, y \in X$. Show that Lipschitz functions are uniformly continuous.

Solution. Given $\varepsilon > 0$, let $\delta = \frac{\varepsilon}{L}$. Then,

$$d_X(x, y) < \delta \implies d_Y(f(x), f(y)) < L \cdot d_X(x, y) < L \cdot \delta = \varepsilon,$$

which is what it means for f to be uniformly continuous. $\square$

Problem 7. Consider the set $\mathbb{Q} \subseteq \mathbb{R}$ with the induced metric

$$d(p, q) = |p - q|.$$

Show that $C = \{q \in \mathbb{Q} : |q| < \sqrt{2}\}$ is closed (in $\mathbb{Q}$) and bounded, but not compact.

Solution. We can write $\mathbb{Q} \setminus C$ as

$$\mathbb{Q} \cap ((-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)).$$

This is an intersection of $\mathbb{Q}$ with a set that is open in $\mathbb{R}$ (because it is a union of open intervals), so $\mathbb{Q} \setminus C$ must be open in $\mathbb{Q}$. Then, the set's complement in $\mathbb{Q}$, namely C is closed in $\mathbb{Q}$.

Moreover, C is bounded because, for any $x, y \in C$ with WLOG $x \geq y$, we have $-\sqrt{2} < y \leq x < \sqrt{2}$, so

$$|x - y| = x - y < \sqrt{2} - (-\sqrt{2}) = 2\sqrt{2},$$

hence the diameter of C is bounded above by $2\sqrt{2}$.

To see that C is not compact, it suffices to prove that it is not sequentially compact (taking the contrapositive of “compact implies sequentially compact”). Indeed, we can take any sequence of rationals less than $\sqrt{2}$ that converges to $\sqrt{2}$ (in $\mathbb{R}$); clearly, this sequence has no subsequential limits in $\mathbb{Q}$. One example of such a sequence is

$$a_n = \frac{1}{10^n} \cdot \lfloor 10^n \sqrt{2} \rfloor,$$

in which $\lfloor x \rfloor$ is the integer part of x , so a_n is the truncation of the decimal representation of $\sqrt{2}$ after n digits past the decimal point (so $a_0 = 1$, $a_1 = 1.4$, $a_2 = 1.41$, and so on). This sequence is always in C and converges to $\sqrt{2}$ because

$$0 \leq \sqrt{2} - a_n = \frac{10^n \sqrt{2} - \lfloor 10^n \sqrt{2} \rfloor}{10^n} < \frac{1}{10^n} \rightarrow 0,$$

as the numerator, the fractional part of $10^n \sqrt{2}$, is always nonnegative but less than 1. $\square$

Problem 8. Let (X, d) be a compact metric space and let $\mathcal{U} = \{U_i \subseteq X : i \in I\}$ be an open cover. Show that there exists $\delta > 0$ such that any subset of X having diameter less than δ is contained in at least one of the open sets in $\mathcal{U}$.

Solution. Since X is compact, it is covered by finitely many of the U_i ; call this finite subcover $U_1, U_2, \dots, U_n$. For each $k = 1, \dots, n$, define the function $f_k: X \rightarrow \mathbb{R}$ as follows:

- If $x \in U_k$, then

$$f_k(x) = \sup\{r : B(x, r) \subseteq U_k\}.$$

Note that U_k being open means that this value is positive.

- If $x \notin U_k$, then $f_k(x) = 0$.

Thus, $f_k(x) = r$ means that any open ball centered at x of radius less than r is contained in U_k (with $r = 0$ meaning no open ball centered at x is a subset of U_k).

Claim 8.1 — Each f_k is a continuous function.

Proof. Let $\varepsilon > 0$ and $x \in X$. Suppose $d(x, y) < \varepsilon$. Then, we have four cases:

- (a) Both x and y are in U_k . Then, letting $r = f_k(x)$,

$$B(y, r - \varepsilon) \subseteq B(x, r) \subseteq B(y, r + \varepsilon)$$

shows that $|f_k(y) - r| < \varepsilon$.

- (b) $x \in U_k$ and y is not. In this case, $y \in B(x, \varepsilon)$ means that $B(x, \varepsilon) \not\subseteq U_k$, so $f_k(x) \leq d(x, y) < \varepsilon$. Thus $|f_k(x) - f_k(y)| = |f_k(x)| < \varepsilon$.

- (c) $y \in U_k$ and x is not. This is entirely symmetric to the case above: we deduce that $f_k(y) < \varepsilon$, so $|f_k(x) - f_k(y)| = |f_k(y)| < \varepsilon$.

- (d) Neither x nor y are in U_k . In this case we have $|f_k(x) - f_k(y)| = 0 < \varepsilon$.

This shows that, if $d(x, y) < \varepsilon$, then

$$d_{\mathbb{R}}(f_k(x), f_k(y)) = |f_k(x) - f_k(y)| < \varepsilon,$$

so f_k is continuous. □

Claim 8.2 — The function

$$F(x) = \max\{f_1(x), \dots, f_n(x)\}$$

is continuous.

Proof. We prove this by induction. The idea is that, for $n = 2$,

$$\max\{f_1(x), f_2(x)\} = \frac{|f_1(x) + f_2(x)| + |f_1(x) - f_2(x)|}{2},$$

which is a sum of a composition of continuous functions and is thus continuous. Then, if the statement holds for some n , it holds for $n + 1$ because

$$\max\{f_1(x), \dots, f_n(x), f_{n+1}(x)\} = \max\{\max\{f_1(x), \dots, f_n(x)\}, f_{n+1}(x)\},$$

and we can use the two-function case again. □

Observe that $F(x) = r$ means that any open ball centered at x with radius less than r will be completely contained within one of the U_k . Moreover, $F(x) > 0$ for all x because each x is in at least one of the U_k , so $f_k(x) > 0$ for at least one k regardless of the value of x . Finally, F is a continuous function whose domain is compact, which means that $F(X)$ is a closed subset of $\mathbb{R}$. Crucially, this means that F attains a minimum value $\delta > 0$.

To finish, let E have diameter less than δ . Then, $E \subseteq B(x, \delta)$ for any $x \in E$, so there exists a $k \in \{1, \dots, n\}$ such that

$$E \subseteq B(x, \delta) \subseteq U_k. \quad \square$$

Problem 9. Prove that a function $f: [0, 1] \rightarrow \mathbb{R}$ is continuous if and only if its graph

$$\Gamma_f = \{(x, f(x)) : x \in [0, 1]\}.$$

is a compact subset of $\mathbb{R}^2$. Give an example of a function $f: [0, 1] \rightarrow \mathbb{R}$ that is discontinuous with graph Γ_f which is closed but not compact.

Solution. Define $\pi: [0, 1] \rightarrow \Gamma_f$ by

$$\pi(x) = (x, f(x)).$$

Claim 9.1 — π is continuous if and only if f is continuous.

Proof. First, suppose f is continuous, and let $c \in [0, 1]$ and $\varepsilon > 0$. By continuity, there exists a δ_1 such that

$$|x - c| < \delta_1 \implies |f(x) - f(c)| < \varepsilon_1$$

where $0 < \varepsilon_1 < \varepsilon$. Of course,

$$|x - c| + |f(x) - f(c)| < \varepsilon_1 + \delta_1,$$

i.e. $\pi(x)$ and $\pi(c)$ have a distance less than $\varepsilon_1 + \delta_1$ in the taxicab metric. Now, we can pick

$$\delta < \min\{\varepsilon - \varepsilon_1, \delta_1\}$$

so that

$$|x - c| < \delta \implies |x - c| + |f(x) - f(c)| < \varepsilon_1 + \delta < \varepsilon_1 + \varepsilon - \varepsilon_1 = \varepsilon.$$

This is exactly what it means for π to be continuous, again recalling that the term that we have bounded above by ε is just the distance between $\pi(x)$ and $\pi(c)$.

For the other direction, if π is continuous, then for each $\varepsilon > 0$ there is a δ such that

$$|x - c| < \delta \implies |f(x) - f(c)| < d_{\text{taxicab}}(\pi(x), \pi(c)) < \varepsilon,$$

so we get continuity of f for free. $\square$

Now, if f is continuous, then π is continuous, meaning $\pi([0, 1]) = \Gamma_f$ is compact because it is the continuous image of a compact set. On the other hand, if Γ_f is compact, consider π^{-1} , which maps $(x, f(x)) \mapsto x$. This is uniformly continuous because for each $\varepsilon > 0$, we can take $\delta = \varepsilon$ to get

$$d_{\text{taxicab}}((x, f(x)), (y, f(y))) = |x - y| + |f(x) - f(y)| < \varepsilon = \delta \implies |x - y| < \varepsilon$$

for any two points $(x, f(x))$ and $(y, f(y))$ in Γ_f . By part (e) of Problem 14, π^{-1} has a continuous inverse (it is bijective because π^{-1} has a inverse), so π is continuous and thus f is continuous.

An example of f such that Γ_f is closed but not compact: take $f(x) = \frac{1}{x}$ for $x \neq 0$ and $f(0) = 0$. The π^{-1} map from earlier is still continuous, so the pre-image $\pi([0, 1]) = \Gamma_f$ is closed because $[0, 1]$ is closed. However, clearly the graph is unbounded because $\frac{1}{x}$ is unbounded on $[0, 1]$, so Γ_f cannot be compact. $\square$

Problem 11. Show that in ℓ^2 the set

$$B = \left\{ \mathbf{x} = (x_1, x_2, x_3, \dots) \in \ell^2 : \sum_{i=1}^{\infty} x_i^2 \leq 1 \right\}$$

is closed and bounded but not compact.

Solution. To show that B is closed, we use a little trick. Denoting the norm of $\mathbf{x}$ as $\|\mathbf{x}\|$, we can see that $f: \ell^2 \rightarrow \mathbb{R}$ given by $f(\mathbf{x}) = \|\mathbf{x}\|$ is continuous because, for each $\varepsilon > 0$, we can take $\delta = \varepsilon$ to get

$$d(\mathbf{x}, \mathbf{y}) < \varepsilon \implies \left| \|\mathbf{x}\| - \|\mathbf{y}\| \right| \leq \|\mathbf{x} - \mathbf{y}\| = d(\mathbf{x}, \mathbf{y}) < \varepsilon,$$

where the inequality of norms follows from the triangle inequality: if WLOG $\|\mathbf{x}\| \geq \|\mathbf{y}\|$, then

$$\|\mathbf{x}\| = d(\mathbf{x}, \mathbf{0}) \leq d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{0}) = \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y}\| \implies \|\mathbf{x}\| - \|\mathbf{y}\| \leq \|\mathbf{x} - \mathbf{y}\|.$$

Since B is by definition the set of all $\mathbf{x}$ such that $\|\mathbf{x}\| \leq 1$, $f(B) \subseteq [0, 1]$, and, for each $\lambda \in [0, 1]$, $\mathbf{x} = (\lambda, 0, 0, \dots) \in B$ means that

$$f(\mathbf{x}) = \sqrt{\lambda^2} = |\lambda| = \lambda \in f(B).$$

That is, $[0, 1] \subseteq f(B)$ too, which proves that $[0, 1] = f(B)$ is a closed set in $\mathbb{R}$. Since f is continuous, the pre-image of a closed set is also closed, so $f^{-1}([0, 1]) = B$ is closed in ℓ^2 .

Additionally, B is bounded because for any $\mathbf{x}$ and $\mathbf{y}$ in B , we can use the fact that $\|\mathbf{x}\| \leq 1$ and $\|\mathbf{y}\| \leq 1$ to say that

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\| = \|\mathbf{x}\| + \|\mathbf{y}\| \leq 2,$$

so any two points in B have a distance at most two, hence $\text{diam } B \leq 2 < +\infty$.

Remark 11.1. We are taking for granted here the vector space structure of ℓ^2 that is established in Problem 12.

Finally, we show that B is not compact by proving that it is not sequentially compact. Indeed, the sequence

$$\mathbf{x}_n = (\underbrace{0, 0, \dots, 0}_{n-1 \text{ zeroes}}, 1, 0, 0, \dots)$$

where the n th coordinate of $\mathbf{x}_n$ is 1 and the rest are zero does not have any convergent subsequences because it doesn't even have any Cauchy subsequences, since

$$d(\mathbf{x}_m, \mathbf{x}_n) = \sqrt{2}$$

for all m and n with $m \neq n$. $\square$

Problem 12. Two infinite vectors $\mathbf{x} = (x_1, x_2, \dots)$ and $\mathbf{y} = (y_1, y_2, \dots)$ can be summed:

$$\mathbf{x} + \mathbf{y} = \{x_1 + y_1, x_2 + y_2, \dots\}$$

and multiplied by a constant $\lambda \in \mathbb{R}$:

$$\lambda \mathbf{x} = (\lambda x_1, \lambda x_2, \dots).$$

Complete the following tasks:

- (a) Show that if $\mathbf{x}$ and $\mathbf{y}$ are in ℓ^2 then so are $\mathbf{x} + \mathbf{y}$ and $\lambda \mathbf{x}$ (i.e. ℓ^2 is a vector space with these addition and scalar multiplication operations).
- (b) Give a definition for what it means for a series

$$\sum_{i=1}^{\infty} \mathbf{x}_i$$

of vectors $\mathbf{x}_1, \mathbf{x}_2, \dots \in \ell^2$ to converge.

- (c) Use your definition to give meaning to infinite linear combinations

$$\mathbf{x} = \sum_{i=1}^{\infty} \lambda_i \mathbf{x}_i.$$

- (d) Show that any vector $\mathbf{x} \in \ell^2$ can be written as an infinite linear combination of the “standard basis vectors” $\mathbf{e}_1 = (1, 0, 0, \dots)$, $\mathbf{e}_2 = (0, 1, 0, \dots)$, $\mathbf{e}_3 = (0, 0, 1, 0, \dots)$, and so on.

Solution. For (a), if $\mathbf{x}$ and $\mathbf{y}$ are in ℓ^2 , then

$$\sum_{i=1}^{\infty} (x_i + y_i)^2 = \sum_{i=1}^{\infty} x_i^2 + y_i^2 + 2x_i y_i = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 + 2 \sum_{i=1}^{\infty} x_i y_i,$$

which converges because $|x_i y_i|$ can be bounded termwise by $\frac{x_i^2 + y_i^2}{2}$ (e.g. by reducing the bound to the equivalent $(|x_i| - |y_i|)^2 \geq 0$). In other words $\mathbf{x} + \mathbf{y}$ is in ℓ^2 because $\sum x_i y_i$ converges absolutely, hence it converges. Additionally,

$$\sum_{i=1}^{\infty} (\lambda x_i)^2 = \lambda^2 \sum_{i=1}^{\infty} x_i^2$$

converges when $\mathbf{x} \in \ell^2$, so $\lambda \mathbf{x} \in \ell^2$ for any real λ too.

For (b) we can define convergence of series in the obvious way: we define

$$\sum_{i=1}^{\infty} \mathbf{x}_i = \lim_{n \rightarrow \infty} \sum_{i=1}^n \mathbf{x}_i$$

to be the limit of the sequence of the partial sums of $\mathbf{x}_i$, if it converges.

For (c) we then define the infinite linear combination to just be the value of the respective infinite sum?

Finally, for (d), I claim that we can write

$$\mathbf{x} = \sum_{i=1}^{\infty} x_i \mathbf{e}_i.$$

Indeed, the n th partial sum of $x_i \mathbf{e}_i$ is $\mathbf{x}_n = (x_1, x_2, \dots, x_n, 0, 0, \dots)$, so

$$\|\mathbf{x} - \mathbf{x}_n\| = \left\| \sum_{i=n+1}^{\infty} x_i \mathbf{e}_i \right\| = \sum_{i=n+1}^{\infty} x_i^2 = \|\mathbf{x}\|^2 - \sum_{i=1}^n x_i^2.$$

Taking the limit as n goes to infinity,

$$\|\mathbf{x}\|^2 - \sum_{i=1}^n x_i^2 \rightarrow 0,$$

so $\|\mathbf{x} - \mathbf{x}_n\|$ becomes arbitrarily small as n grows large, meaning $\mathbf{x}_n$ converges to $\mathbf{x}$. $\square$

Problem 13. A function $\varphi: \ell^2 \rightarrow \mathbb{R}$ is called **linear** if

$$\varphi(\mathbf{x} + \mathbf{y}) = \varphi(\mathbf{x}) + \varphi(\mathbf{y}) \quad \text{and} \quad \varphi(\lambda \mathbf{x}) = \lambda \varphi(\mathbf{x})$$

for all $\mathbf{x}, \mathbf{y} \in \ell^2$. Show that if $\varphi: \ell^2 \rightarrow \mathbb{R}$ is linear then the following statements are equivalent:

- (i) φ is continuous;
- (ii) φ is uniformly continuous;
- (iii) φ is Lipschitz; and
- (iv) φ has bounded norm, meaning

$$\sup \left\{ |\varphi(\mathbf{x})| : \mathbf{x} \in \ell^2 \text{ such that } \sum_{i=1}^{\infty} x_i^2 \leq 1 \right\} < +\infty.$$

Solution. We know that

$$(iii) \implies (ii) \implies (i).$$

We will prove additionally that (i) implies (iv) and (iv) implies (iii).

First we show (i) implies (iv). Suppose for the sake of contradiction that φ does not have a bounded norm.

Claim 13.1 — $|\varphi(\mathbf{x})|$ is unbounded on every open ball centered at $\mathbf{0}$.

Proof. Obviously φ being bounded on the closed unit ball means it is bounded on the open unit ball. Define

$$L = \sup\{|\varphi(\mathbf{x})| : \|\mathbf{x}\| < 1\}$$

and let $\varepsilon > 0$. Then, we have

$$\sup\{|\varphi(\mathbf{y})| : \|\mathbf{y}\| < \varepsilon\} = \sup\{|\varphi(\mathbf{x} \cdot \varepsilon)| : \|\mathbf{x}\| < 1\} = \sup\{\varepsilon \cdot |\varphi(\mathbf{x})| : \|\mathbf{x}\| < 1\} = L\varepsilon,$$

so φ is bounded on the open ball of radius ε for any $\varepsilon > 0$, as desired. $\square$

On the other hand, let U be the pre-image of the open interval $(-1, 1)$. Since φ is continuous, U is open in ℓ^2 . Note that

$$\varphi(\mathbf{0}) = \varphi(2 \cdot \mathbf{0}) = 2 \cdot \varphi(\mathbf{0}),$$

so $\varphi(\mathbf{0}) = 0$, hence $\mathbf{0} \in U$. In particular, U also contains some open ball B containing $\mathbf{0}$. That is, for all $\mathbf{x} \in B$, $\varphi(\mathbf{x}) < 1$, which means φ is bounded on B , contradicting the claim.

To show that (iv) implies (iii), we need to prove that if φ has a bounded norm then there exists a constant L such that

$$|\varphi(\mathbf{x} - \mathbf{y})| = |\varphi(\mathbf{x}) - \varphi(\mathbf{y})| \leq L \cdot \|\mathbf{x} - \mathbf{y}\|.$$

This is obvious when $\mathbf{x} = \mathbf{y}$, so assume otherwise. Then, letting $\mathbf{u} = \mathbf{x} - \mathbf{y}$, we want to show that the function $f: \ell^2 \setminus \mathbf{0} \rightarrow \mathbb{R}$

$$f(\mathbf{u}) = \frac{|\varphi(\mathbf{u})|}{\|\mathbf{u}\|}$$

is bounded as $\mathbf{u}$ varies across the domain of f . Noting that $f(\lambda\mathbf{u}) = f(\mathbf{u})$ for any nonzero real λ , we can assume without loss of generality that $\|\mathbf{u}\| = 1$. The result is then immediate by our assumption that φ has bounded norm, since $f(\mathbf{u}) = |\varphi(\mathbf{u})|$ is bounded on $\{\mathbf{u} \in \ell^2 : \|\mathbf{u}\| = 1\}$. $\square$

Problem 14. Two metric spaces X and Y are **topologically equivalent** or **homeomorphic** if there exists a **homeomorphism** between them, which is a bijection $f: X \rightarrow Y$ such that both f and f^{-1} are continuous. Complete the following tasks:

- Show that the interval $(0, 1)$ is homeomorphic to $\mathbb{R}$.
- Explain why, if X and Y are topologically equivalent metric spaces, then X is *connected* if and only if Y is *connected*.
- Prove that the above statement is true if we substitute the word *connected* with *path connected* or *compact*.
- Show an example of a continuous bijective function $f: X \rightarrow Y$ that is not a homeomorphism.
- Prove that if X is compact then any continuous bijective function $f: X \rightarrow Y$ is a homeomorphism.

We say that X and Y are **metrically equivalent** or **isometric** if there exists an **isometry** $f: X \rightarrow Y$, which is a bijective function that preserves distance, i.e. for all $x, y \in X$,

$$d_Y(f(x), f(y)) = d_X(x, y).$$

Show that

- if $f: X \rightarrow Y$ is an isometry then so is f^{-1} .
- there exists a pair of metric spaces that are homeomorphic but not isometric.

Solution. First, for (a), it is easy to check that topological equivalence is an equivalence relation. Moreover, any two open intervals are homeomorphic because $g: (a, b) \rightarrow (c, d)$ given by

$$g(x) = c + \frac{x - a}{b - a} \cdot (d - c)$$

is easily checked to be a homeomorphism (it is continuous; it is bijective because it is monotonic and surjective; and its inverse is continuous because the inverse is also a linear polynomial in x).

Now, consider the function $f: (-1, 1) \rightarrow \mathbb{R}$ given by

$$f(x) = \frac{x}{\sqrt{1-x^2}}.$$

This is clearly continuous, and we can check that it has an inverse

$$f^{-1}(x) = \frac{x}{\sqrt{1+x^2}}.$$

The existence of the inverse implies that f is bijective, and we can see that f^{-1} is continuous. Thus $\mathbb{R}$ is homeomorphic to $(-1, 1)$, which is homeomorphic to $(0, 1)$.

For (b) and (c), we use the following fact (which justifies the phrase “topologically equivalent”):

Theorem. Let $f: X \rightarrow Y$ be a homeomorphism. Then $U \subseteq X$ is open if and only if $f(U) \subseteq Y$ is open.

Proof. First, if $f(U)$ is open in Y , then continuity says that $f^{-1}(f(U)) = U$ is open in X , where $f^{-1}(f(U)) = U$ because f is bijective.

Similarly, let $g = f^{-1}$; then, if U is open in X , continuity of g shows that $g^{-1}(U) = f(U)$ is open in Y . $\square$

In each case, we only prove that Y has the property if X has it (the other direction follows by symmetry).

For connectedness, we prove equivalently that X is disconnected if and only if Y is disconnected. Indeed, suppose that X is a union of disjoint nonempty open sets A and B . Then,

$$f(A) = \{f(x) : x \in A\}, \quad f(B) = \{f(x) : x \in B\}$$

are open in Y by the theorem and are nonempty. The union of these sets is

$$f(A) \cup f(B) = \{f(x) : x \in A \text{ or } x \in B\} = \{f(x) : x \in A \cup B\} = \{f(x) : x \in X\} = Y$$

with the last equality following from f being a bijection. Similarly the intersection of the sets is

$$f(A) \cap f(B) = \{f(x) : x \in A \text{ and } x \in B\} = \{f(x) : x \in A \cap B\} = \{f(x) : x \in \emptyset\} = \emptyset.$$

Thus $f(A)$ and $f(B)$ are disjoint nonempty open sets whose union is Y , i.e. Y is disconnected.

For path connectedness, the idea is that we can push a path from one space to another because composition of continuous functions is continuous. Specifically, if X is path connected, then let a and b be points in Y . We can construct a path $\gamma: [0, 1] \rightarrow X$ between $f^{-1}(a)$ and $f^{-1}(b)$ so that $f \circ \gamma$ is continuous and has $f(\gamma(0)) = a$ and $f(\gamma(1)) = b$, i.e. $f \circ \gamma$ is a path between a and b . The other direction follows by the exact same logic.

For compactness, let $\mathcal{V} = \{V_i : i \in I\}$ be an open cover of Y . Then,

$$\mathcal{U} = \{U_i : i \in I\}$$

where $U_i = f^{-1}(V_i)$ is an open cover of X because

$$\bigcup_{i \in I} U_i = \bigcup_{i \in I} \{f^{-1}(y) : y \in V_i\} = \left\{ f^{-1}(y) : y \in \bigcup_{i \in I} V_i \right\} = \{f^{-1}(y) : y \in Y\} = X$$

with the last equality following from f being bijective. Then, we can find a finite subcover of X , say $U_{i_1}, \dots, U_{i_n}$. Since f is bijective, $f(U_{i_k}) = V_{i_k}$, and the union of this subcollection of $\mathcal{V}$ is

$$\bigcup_{k=1}^n \{f(x) : x \in U_{i_k}\} = \left\{ f(x) : x \in \bigcup_{k=1}^n U_{i_k} \right\} = \{f(x) : x \in X\} = Y,$$

i.e. we have extracted a finite subcover of Y from $\mathcal{V}$.

For (d), let $X = \mathbb{N}$, $Y = \mathbb{Q}$, and let $f : X \rightarrow Y$ be any bijection between $\mathbb{N}$ and $\mathbb{Q}$.

On one hand, f is continuous because every point in $\mathbb{N}$ is an isolated point: to show that f is continuous at $n \in \mathbb{N}$, we can take $\delta < 1$ so that $B_X(n, \delta) = \{n\}$ (because the only integer within δ of n is n), and then

$$|x - n| < \delta \iff x = n \implies |f(x) - f(n)| = 0 < \varepsilon$$

for any $\varepsilon > 0$.

On the other hand, f^{-1} is not continuous at any point of $\mathbb{Q}$. Taking $q \in \mathbb{Q}$, we know because f^{-1} is bijective that if $x \neq q$ is rational then $f(q)$ and $f(x)$ are distinct elements of $\mathbb{N}$, which differ by at least 1. That is, regardless of the value of $|q - x|$, the inequality $|f(q) - f(x)| \geq 1$. Thus, taking $q_n = q + \frac{1}{n}$, we have

$$\lim_{n \rightarrow \infty} q_n = q,$$

but the limit of $f(q_n)$ is not equal to $f(q)$ because $|f(q) - f(q_n)| \geq 1$ for all n , which finishes the proof of discontinuity.

For (e), to show that f^{-1} is continuous, we can show that f maps closed sets to closed sets. This is pretty simple: if E is a closed subset of X , then it is also compact. Then, $f(E)$ is also compact, hence closed!

For (f), we have that, given x and y in Y , we can write $x = f(u)$, $y = f(v)$ where u, v are in X . Then

$$d_Y(x, y) = d_Y(f(u), f(v)) = d_X(u, v) = d_X(f^{-1}(x), f^{-1}(y)).$$

For (g), observe that $(0, 1)$ and $\mathbb{R}$ are homeomorphic (as shown in (a)), but they are not isometric. Indeed, if they were isometric, then there exist α and β in $(0, 1)$ such that $f(\alpha) = 0$ and $f(\beta) = 2$, so that

$$2 = |f(\alpha) - f(\beta)| = |\alpha - \beta|$$

which is a contradiction because $\text{diam}((0, 1)) = 1 < 2$. □

Problem 15. Let $\mathbf{a} = (a_1, a_2, \dots)$ be an element of ℓ^2 . Show that the function $\varphi_{\mathbf{a}}: \ell^2 \rightarrow \mathbb{R}$ defined by

$$\varphi_{\mathbf{a}}(\mathbf{x}) = \sum_{i=1}^{\infty} a_i x_i$$

is well-defined (i.e. the series on the right converges) and is linear and continuous.

In 1908, the Hungarian mathematician Frigyes Riesz proved that any continuous linear function $\ell^2 \rightarrow \mathbb{R}$ is expressible in this way — try proving this!

Solution. The series converges because it can be bounded termwise by

$$|a_i x_i| \leq \frac{a_i^2 + x_i^2}{2}$$

(by expanding and reducing to the equivalent $(|a_i| - |x_i|)^2 \geq 0$). This shows that $\sum a_i x_i$ converges absolutely, so it converges.

It is linear because

$$\varphi_{\mathbf{a}}(\mathbf{x}) + \varphi_{\mathbf{a}}(\mathbf{y}) = \sum_{i=1}^{\infty} a_i x_i + \sum_{i=1}^{\infty} a_i y_i = \sum_{i=1}^{\infty} a_i (x_i + y_i) = \varphi_{\mathbf{a}}(\mathbf{x} + \mathbf{y}).$$

To finish, we have

$$\varphi_{\mathbf{a}}(\mathbf{x}) \leq \frac{\|\mathbf{a}\|^2 + \|\mathbf{x}\|^2}{2},$$

so, when $\mathbf{x}$ has norm at most 1,

$$\varphi_{\mathbf{a}}(\mathbf{x}) \leq \frac{\|\mathbf{a}\|^2 + 1}{2},$$

hence $\varphi_{\mathbf{a}}$ is bounded on the closed unit ball; Problem 13 then allows us to conclude that $\varphi_{\mathbf{a}}$ is continuous. $\square$